Non-local Games and Quantum Foundations

Chapter 3

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Outline

3.1 Basic Definitions

3.2 The CHSH Game

3.3 Self-testing a Bell Pair

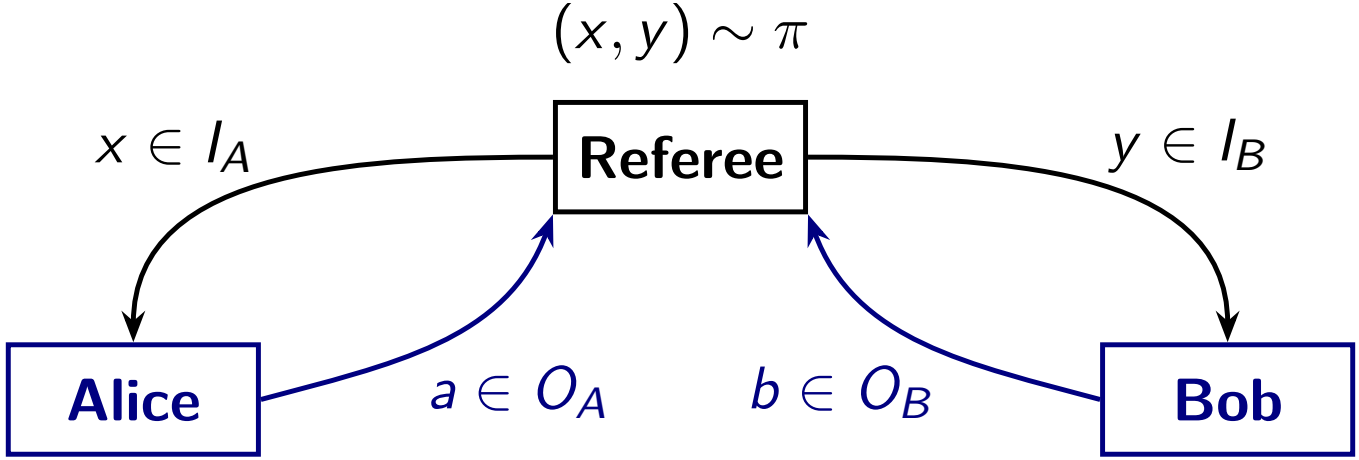
3.4 The Mermin–Peres Magic Square

3.5 Quantum Contextuality

Basic Definitions

A non-local game

A game is $G = (I_A, I_B, O_A, O_B, \pi, V)$: the question sets I_A, I_B and answer sets O_A, O_B are finite and nonempty, π is a distribution on $I_A \times I_B$, and $V(a, b, x, y) \in \{0, 1\}$.



One round: the players win if $V(a, b, x, y) = 1$.
The players may share a strategy and resources beforehand, independently of the questions.
No communication after receiving the questions.

Correlations, success probability, and value

A strategy produces conditional probabilities

$$p(a, b \mid x, y) \geq 0, \qquad \sum_{a,b} p(a, b \mid x, y) = 1.$$

Its success probability is

$$\boxed{\omega(G, p) = \sum_{x,y} \pi(x, y) \sum_{a,b} V(a, b, x, y) p(a, b \mid x, y).}$$

A strategy is perfect if $\omega(G, p) = 1$.

The allowed resources determine the value

For a nonempty class $\mathcal{S}$ of strategies, set

$$\omega(G, \mathcal{S}) = \sup_{p \in \mathcal{S}} \omega(G, p).$$

We will compare classical strategies with quantum strategies.

Deterministic strategies

Alice and Bob choose functions

$$f\colon I_A \to O_A, \qquad g\colon I_B \to O_B.$$

On questions x, y, their answers are $a = f(x)$ and $b = g(y)$. Thus

$$p(a, b \mid x, y) = \begin{cases} 1, & a = f(x) \text{ and } b = g(y), \\ 0, & \text{otherwise.} \end{cases}$$

Their success probability is simply

$$\omega(G; f, g) = \sum_{x,y} \pi(x, y) V(f(x), g(y), x, y).$$

There are only finitely many pairs (f, g).

Classical strategies

Alice and Bob share a discrete random variable $\lambda \in \Lambda$, independent of the questions, with probability distribution

$$p_\Lambda(\lambda) \geq 0, \qquad \sum_{\lambda \in \Lambda} p_\Lambda(\lambda) = 1.$$

Given λ and their questions, they independently answer using local probability distributions $p_A(a \mid x, \lambda)$ and $p_B(b \mid y, \lambda)$. Thus a classical (or randomized) strategy has correlations

$$\boxed{p(a, b \mid x, y) = \sum_{\lambda \in \Lambda} p_\Lambda(\lambda) p_A(a \mid x, \lambda) p_B(b \mid y, \lambda).}$$

The classical value is

$$\omega_c(G) = \sup_{p \text{ classical}} \omega(G, p).$$

Quantum strategies

Alice and Bob share a state ρ_{AB} on $\mathcal{H}_A \otimes \mathcal{H}_B$, independent of the questions. For question x, Alice uses a POVM $\{\mu_A^x(a)\}_a$. For question y, Bob uses $\{\mu_B^y(b)\}_b$.

$$\boxed{p(a,b \mid x,y) = \mathrm{tr}\big[(\mu_A^x(a) \otimes \mu_B^y(b))\rho_{AB}\big].}$$

These probabilities are nonnegative: for $F, \rho \geq 0$, $\mathrm{tr}(F\rho) = \mathrm{tr}(\rho^{1/2} F \rho^{1/2}) \geq 0$. Also,

$$\sum_{a,b} p(a,b \mid x,y) = \mathrm{tr}[(I_A \otimes I_B)\rho_{AB}] = 1.$$

The quantum value $\omega_q(G)$ is the supremum over all such strategies, allowing arbitrary finite local dimensions.

Randomization and separable states

Randomized strategies

Suppose a randomized strategy p wins a non-local game G with success probability $\omega(G, p)$. Then there exists a deterministic strategy for G with success probability at least $\omega(G, p)$.

Separable states

Suppose a quantum strategy p using a separable state ρ_{AB} wins a non-local game G with success probability $\omega(G, p)$. Then there exists a randomized, and hence also a deterministic, strategy for G with success probability at least $\omega(G, p)$.

Proofs are left for homework.

The CHSH Game

The CHSH game

Questions and answers are bits: $I_A = I_B = O_A = O_B = \{0, 1\}$.
Each question pair has probability $1/4$. The winning condition is

$$\boxed{a \oplus b = xy,}$$

where $\oplus$ denotes addition modulo two.

x	y	xy	Required answers
0	0	0	Same
0	1	0	Same
1	0	0	Same
1	1	1	Different

The classical CHSH value

Randomization cannot improve the optimal winning probability. Consider deterministic answers $a = f(x)$ and $b = g(y)$. Winning on all four question pairs would require

$$f(0) \oplus g(0) = 0, \quad f(0) \oplus g(1) = 0,$$
$$f(1) \oplus g(0) = 0, \quad f(1) \oplus g(1) = 1.$$

Add the four equations modulo two. Each function value occurs twice on the left, giving $0 = 1$. Thus every deterministic strategy loses on at least one pair:

$$\omega_c(G) \leq \frac{3}{4}.$$

Always answering $a = b = 0$ wins on the first three pairs, so

$$\boxed{\omega_c(G) = \frac{3}{4}.}$$

□

Specifying a quantum CHSH strategy

To construct a quantum strategy, we must specify:

1. **Local Hilbert spaces.** We use one qubit each:

$$\mathcal{H}_A = \mathcal{H}_B = \mathbb{C}^2.$$

2. **The shared state.** We use the maximally entangled state

$$|\Phi^+\rangle = \frac{|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle}{\sqrt{2}}, \qquad \rho_{AB} = |\Phi^+\rangle\langle\Phi^+|.$$

3. **Local POVMs.** For each question $x, y \in \{0, 1\}$, we must choose binary measurements

$$\{\mu_A^x(0), \mu_A^x(1)\}, \qquad \{\mu_B^y(0), \mu_B^y(1)\}.$$

We will use projective measurements, choosing their bases using Bloch vectors.

The observable of a Bloch vector

The Z- and X-basis measurements, with outcomes $+1, -1$, correspond to the observables

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|, \qquad X = |+\rangle\langle +| - |-\rangle\langle -|.$$

More generally, let $\vec{r} = (r_x, r_y, r_z)$ be a unit Bloch vector. The antipodal points $\vec{r}$ and $-\vec{r}$ represent orthogonal pure states, with projectors

$$P_\pm = \rho(\pm\vec{r}) = \frac{1}{2}(I \pm (r_x X + r_y Y + r_z Z)).$$

Assign outcome $+1$ to P_+ and -1 to P_-. The observable is

$$\begin{aligned} O(\vec{r}) &= (+1)P_+ + (-1)P_- = P_+ - P_- \\ &= r_x X + r_y Y + r_z Z = \begin{pmatrix} r_z & r_x - ir_y \\ r_x + ir_y & -r_z \end{pmatrix}. \end{aligned}$$

Local observables and joint measurements

Let Alice's and Bob's observables have spectral decompositions

$$O_A = \sum_\alpha \alpha P_\alpha, \qquad O_B = \sum_\beta \beta Q_\beta.$$

The joint outcome (α, β) corresponds to the measurement operator

$$R_{\alpha\beta} = P_\alpha \otimes Q_\beta.$$

These operators are Hermitian and satisfy

$$R_{\alpha\beta} R_{\alpha'\beta'} = \delta_{\alpha\alpha'} \delta_{\beta\beta'} R_{\alpha\beta},$$
$$\sum_{\alpha,\beta} R_{\alpha\beta} = \Big(\sum_\alpha P_\alpha\Big) \otimes \Big(\sum_\beta Q_\beta\Big) = I_A \otimes I_B.$$

Thus they form a projective measurement. By the Born rule,

$$\mathbb{P}(\alpha, \beta) = \mathrm{tr}[(P_\alpha \otimes Q_\beta)\rho_{AB}].$$

Tensor products and outcome products

Reporting the product $\alpha\beta$ of the local outcomes gives the observable

$$\sum_{\alpha,\beta} \alpha\beta\, P_\alpha \otimes Q_\beta = \Big(\sum_\alpha \alpha P_\alpha\Big) \otimes \Big(\sum_\beta \beta Q_\beta\Big) = O_A \otimes O_B.$$

By linearity of the trace, its expectation is

$$\begin{aligned} \mathbb{E}[\alpha\beta] &= \sum_{\alpha,\beta} \alpha\beta \ \mathrm{tr}[(P_\alpha \otimes Q_\beta)\rho_{AB}] \\ &= \boxed{\mathrm{tr}[(O_A \otimes O_B)\rho_{AB}].} \end{aligned}$$

For a pure shared state $\rho_{AB} = |\psi\rangle\langle\psi|$, this becomes

$$\mathbb{E}[\alpha\beta] = \langle\psi| O_A \otimes O_B |\psi\rangle.$$

Example: $X \otimes Z$

Alice measures X and Bob measures Z on $|\Phi^+\rangle$.

$$|\Phi^+\rangle = \frac{1}{2}(|+\rangle \otimes |0\rangle + |+\rangle \otimes |1\rangle + |-\rangle \otimes |0\rangle - |-\rangle \otimes |1\rangle),$$

Each squared amplitude is $1/4$:

Eigenvector	Pair (α, β)	Product $\alpha\beta$	Probability
$\|+\rangle \otimes \|0\rangle$	$(+1, +1)$	$+1$	$1/4$
$\|+\rangle \otimes \|1\rangle$	$(+1, -1)$	-1	$1/4$
$\|-\rangle \otimes \|0\rangle$	$(-1, +1)$	-1	$1/4$
$\|-\rangle \otimes \|1\rangle$	$(-1, -1)$	$+1$	$1/4$

The observable $X \otimes Z$ reports the product: $+1$ or -1, each with probability $1/2$. Thus

$$\langle\Phi^+|X \otimes Z|\Phi^+\rangle = \mathbb{E}[\alpha\beta] = \frac{1}{2} - \frac{1}{2} = 0.$$

Our quantum CHSH strategy

We share the maximally entangled state

$$|\Phi^+\rangle = \frac{|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle}{\sqrt{2}}.$$

We need four observables: Alice has A_0, A_1, and Bob has B_0, B_1.
On questions x, y, they measure A_x and B_y.
All four measurements will be projective, with unit Bloch vectors in the xz-plane:

$$\vec{r} = (r_x, 0, r_z), \qquad r_x^2 + r_z^2 = 1, \qquad O(\vec{r}) = r_x X + r_z Z.$$

Encoding the CHSH answers

The answer bits $a, b \in \{0, 1\}$ correspond to measurement signs

$$\alpha = (-1)^a, \qquad \beta = (-1)^b,$$

$$\boxed{0 \longleftrightarrow +1, \qquad 1 \longleftrightarrow -1.}$$

An identity for the maximally entangled state

Let $|\Phi^+\rangle = (|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)/\sqrt{2}$. For arbitrary 2×2 matrices M, N,

$$\boxed{\langle \Phi^+ | M \otimes N | \Phi^+ \rangle = \frac{1}{2} \operatorname{tr}[M^{\mathsf{T}} N].}$$

Here M^{T} is the transpose in the computational basis.

Proof. Write $M_{ij} = \langle i|M|j\rangle$ and $N_{ij} = \langle i|N|j\rangle$. Expanding the Bell state gives

$$\begin{aligned}\langle \Phi^+ | M \otimes N | \Phi^+ \rangle &= \frac{1}{2} \sum_{i,j=0}^{1} M_{ij} N_{ij} \\ &= \frac{1}{2} \sum_{j=0}^{1} (M^{\mathsf{T}} N)_{jj} = \frac{1}{2} \operatorname{tr}[M^{\mathsf{T}} N].\end{aligned}$$

□

Correlations are cosines

Write $\langle T \rangle = \langle \Phi^+ | T | \Phi^+ \rangle$. For unit Bloch vectors $\vec{r}, \vec{s}$ in the xz-plane, the observables $O(\vec{r}), O(\vec{s})$ are real symmetric. The Bell-state trace identity therefore gives

$$\begin{aligned} \langle O(\vec{r}) \otimes O(\vec{s}) \rangle &= \frac{1}{2} \operatorname{tr}[O(\vec{r})^{\mathsf{T}} O(\vec{s})] \\ &= \frac{1}{2} \operatorname{tr}[(r_x X + r_z Z)(s_x X + s_z Z)]. \end{aligned}$$

We have $X^2 = Z^2 = I$ and $XZ = -ZX$. Using cyclicity of the trace,

$$\operatorname{tr}(X^2) = \operatorname{tr}(Z^2) = 2, \qquad \operatorname{tr}(XZ) = \operatorname{tr}(ZX) = 0.$$

Expanding the trace now gives

$$\boxed{\langle O(\vec{r}) \otimes O(\vec{s}) \rangle = r_x s_x + r_z s_z = \vec{r} \cdot \vec{s} = \cos \delta,}$$

where δ is the angle between the two unit Bloch vectors.

The four winning events

We win when the answers are the same for $(0,0),(0,1),(1,0)$, and different for $(1,1)$. The four question pairs are equally likely, so

$$\mathbb{P}(\text{win}) = \frac{1}{4}\big[\mathbb{P}(a=b \mid 0,0) + \mathbb{P}(a=b \mid 0,1) + \mathbb{P}(a=b \mid 1,0) + \mathbb{P}(a\neq b \mid 1,1)\big].$$

The product of the signs $\alpha=(-1)^a$, $\beta=(-1)^b$ is $+1$ for the same answers and -1 for different answers. Taking its expectation gives

$$\begin{aligned}\langle A_x \otimes B_y\rangle &= \mathbb{E}[\alpha\beta \mid x,y] \\ &= \mathbb{P}(a=b \mid x,y) - \mathbb{P}(a\neq b \mid x,y) \\ &= 2\mathbb{P}(a=b \mid x,y) - 1.\end{aligned}$$

The last equality uses $\mathbb{P}(a=b \mid x,y) + \mathbb{P}(a\neq b \mid x,y) = 1$.

Substituting the observable expectations

Solving for the two probabilities gives

$$\mathbb{P}(a = b \mid x, y) = \frac{1 + \langle A_x \otimes B_y \rangle}{2},$$
$$\mathbb{P}(a \neq b \mid x, y) = \frac{1 - \langle A_x \otimes B_y \rangle}{2}.$$

Substitute these into the four terms of the winning probability:

$$\mathbb{P}(\text{win}) = \frac{1}{4}\Big[\frac{1 + \langle A_0 \otimes B_0 \rangle}{2} + \frac{1 + \langle A_0 \otimes B_1 \rangle}{2} + \frac{1 + \langle A_1 \otimes B_0 \rangle}{2} + \frac{1 - \langle A_1 \otimes B_1 \rangle}{2}\Big].$$

The minus sign in the last term comes from requiring different answers on questions $(1, 1)$.

The CHSH operator

Collecting the constant terms gives

$$\mathbb{P}(\text{win}) = \frac{1}{2} + \frac{1}{8}\big[\langle A_0 \otimes B_0\rangle + \langle A_0 \otimes B_1\rangle + \langle A_1 \otimes B_0\rangle - \langle A_1 \otimes B_1\rangle\big].$$

By linearity, the bracket is the expectation of the CHSH operator

$$\begin{aligned} M_{AB} &= A_0 \otimes B_0 + A_0 \otimes B_1 + A_1 \otimes B_0 - A_1 \otimes B_1 \\ &= A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1). \end{aligned}$$

Writing $S = \langle M_{AB}\rangle$, we obtain

$$\boxed{\mathbb{P}(\text{win}) = \frac{1}{2} + \frac{1}{8}\langle M_{AB}\rangle = \frac{1}{2} + \frac{S}{8}.}$$

Our goal is now to maximize S over the four observables.

Choosing Alice's directions

Write the four unit Bloch vectors as $\vec{a}_0, \vec{a}_1$ for Alice and $\vec{b}_0, \vec{b}_1$ for Bob. Since $\langle A_x \otimes B_y \rangle = \vec{a}_x \cdot \vec{b}_y$,

$$\begin{aligned} S &= \vec{a}_0 \cdot \vec{b}_0 + \vec{a}_0 \cdot \vec{b}_1 + \vec{a}_1 \cdot \vec{b}_0 - \vec{a}_1 \cdot \vec{b}_1 \\ &= \vec{a}_0 \cdot (\vec{b}_0 + \vec{b}_1) + \vec{a}_1 \cdot (\vec{b}_0 - \vec{b}_1). \end{aligned}$$

For a unit vector $\vec{u}$, Cauchy–Schwarz gives $\vec{u} \cdot \vec{v} \leq \|\vec{v}\|$, with equality when $\vec{u}$ points along $\vec{v}$. Thus, for fixed Bob directions, Alice chooses

$$\vec{a}_0 \text{ along } \vec{b}_0 + \vec{b}_1, \qquad \vec{a}_1 \text{ along } \vec{b}_0 - \vec{b}_1.$$

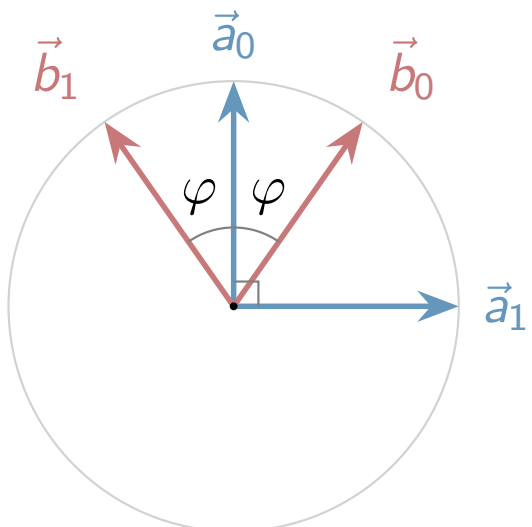
If either vector is zero, that term is zero for every Alice direction. Therefore

$$\boxed{\max_{\vec{a}_0, \vec{a}_1} S = \|\vec{b}_0 + \vec{b}_1\| + \|\vec{b}_0 - \vec{b}_1\|.}$$

One angle determines the geometry

Let the angle between Bob's vectors be 2φ, with $0 \leq \varphi \leq \pi/2$.
Their sum and difference are perpendicular:

$$\begin{aligned}(\vec{b}_0 + \vec{b}_1) \cdot (\vec{b}_0 - \vec{b}_1) &= \|\vec{b}_0\|^2 - \vec{b}_0 \cdot \vec{b}_1 + \vec{b}_1 \cdot \vec{b}_0 - \|\vec{b}_1\|^2 \\ &= \|\vec{b}_0\|^2 - \|\vec{b}_1\|^2 = 1 - 1 = 0.\end{aligned}$$



Since $\vec{b}_0 \cdot \vec{b}_1 = \cos(2\varphi)$,

$$\begin{aligned}\|\vec{b}_0 + \vec{b}_1\|^2 &= 2 + 2\cos(2\varphi) \\ &= 4\cos^2\varphi, \\ \|\vec{b}_0 - \vec{b}_1\|^2 &= 2 - 2\cos(2\varphi) \\ &= 4\sin^2\varphi.\end{aligned}$$

Therefore their lengths are $2\cos\varphi$ and $2\sin\varphi$.

Optimizing the angle

For our Bell-state strategy in the xz-plane, the best Alice choices give

$$S_{\max}(\varphi) = 2(\cos\varphi + \sin\varphi), \qquad 0 \le \varphi \le \frac{\pi}{2}.$$

Using $\cos^2\varphi + \sin^2\varphi = 1$,

$$(\cos\varphi + \sin\varphi)^2 = 2 - (\cos\varphi - \sin\varphi)^2 \le 2.$$

Thus $S_{\max}(\varphi) \le 2\sqrt{2}$. Equality holds precisely when

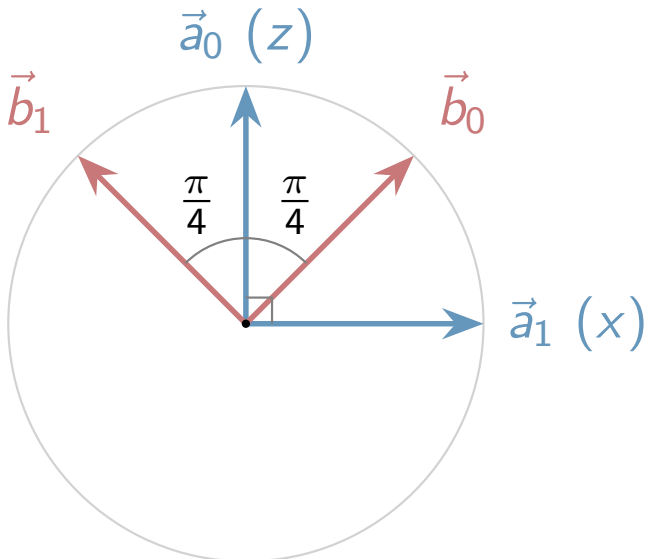
$$\cos\varphi = \sin\varphi, \qquad \text{that is,} \qquad \boxed{\varphi = \frac{\pi}{4}.}$$

The optimal geometry

Bob's directions are $2\varphi = \pi/2$ apart. Alice's directions point along their sum and their difference.

The four optimal observables

Choose Alice's perpendicular directions to be the z- and x-axes. Bob's directions are at $\pm\pi/4$ from the z-axis.



Using $O(\vec{r}) = r_x X + r_z Z$,

$$A_0 = Z, \qquad A_1 = X,$$

$$B_0 = \frac{Z + X}{\sqrt{2}},$$

$$B_1 = \frac{Z - X}{\sqrt{2}}.$$

The three angles $(\vec{a}_0, \vec{b}_0)$, $(\vec{a}_0, \vec{b}_1)$, $(\vec{a}_1, \vec{b}_0)$ are $\pi/4$; the angle $(\vec{a}_1, \vec{b}_1)$ is $3\pi/4$.

The resulting success probability

For our four Bloch directions, $\langle A_x \otimes B_y \rangle = \cos \delta_{xy}$:

Questions (x, y)	Angle δ_{xy}	$\langle A_x \otimes B_y \rangle$
$(0, 0)$	$\pi/4$	$1/\sqrt{2}$
$(0, 1)$	$\pi/4$	$1/\sqrt{2}$
$(1, 0)$	$\pi/4$	$1/\sqrt{2}$
$(1, 1)$	$3\pi/4$	$-1/\sqrt{2}$

$$\begin{aligned}\mathbb{P}(\text{win}) &= \frac{1}{2} + \frac{1}{8}\langle M_{AB} \rangle \\ &= \frac{1}{2} + \frac{1}{8}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - \frac{-1}{\sqrt{2}}\right) = \frac{1}{2} + \frac{1}{2\sqrt{2}}.\end{aligned}$$

$$\boxed{\omega_q(G) \geq \mathbb{P}(\text{win}) = \frac{1}{2} + \frac{1}{2\sqrt{2}} \approx 0.8536.}$$

Bell's theorem

Quantum correlations cannot always be reproduced by local hidden-variable models. In the language of non-local games:

$$\boxed{\text{There exists a game } G \text{ with } \omega_c(G) < \omega_q(G).}$$

Proof. Take the CHSH game. The bounds already proved give

$$\omega_c(G) = \frac{3}{4} < \frac{1}{2} + \frac{1}{2\sqrt{2}} \leq \omega_q(G).$$

□

The upper bound: projective measurements

We must prove that every quantum strategy satisfies

$$\mathbb{P}(\text{win}) \leq \frac{1}{2} + \frac{1}{2\sqrt{2}}.$$

A measurement reduction (proved in a later chapter)
Any finite family of POVMs admits projective measurements on a larger space, using one fixed embedding of the state. Applied locally, this preserves all joint probabilities.
For a binary projective measurement, the observable $R = P_0 - P_1$ is Hermitian. Since $P_0 + P_1 = I$ and $P_0 P_1 = P_1 P_0 = 0$,

$$R^2 = P_0^2 + P_1^2 - P_0 P_1 - P_1 P_0 = P_0 + P_1 = I.$$

Thus A_x, B_y are Hermitian, with $(A_x)^2 = I_A$ and $(B_y)^2 = I_B$.

Reduction to a pure shared state

Fix the measurements and write the spectral decomposition

$$\rho_{AB} = \sum_j p_j |\psi_j\rangle\langle\psi_j|, \qquad p_j \geq 0, \quad \sum_j p_j = 1, \quad \||\psi_j\rangle\| = 1.$$

By linearity of the trace, each joint probability satisfies

$$\begin{aligned} p_\rho(a, b \mid x, y) &= \mathrm{tr}[(\mu_A^x(a) \otimes \mu_B^y(b))\rho_{AB}] \\ &= \sum_j p_j \langle\psi_j|\mu_A^x(a) \otimes \mu_B^y(b)|\psi_j\rangle \\ &= \sum_j p_j p_{\psi_j}(a, b \mid x, y). \end{aligned}$$

The winning probability is a weighted sum of these probabilities, so

$$\boxed{\mathbb{P}_\rho(\mathrm{win}) = \sum_j p_j \mathbb{P}_{\psi_j}(\mathrm{win}).}$$

Thus it suffices to prove the bound for every pure state.

Cauchy–Schwarz reduces the problem

Let $|\psi\rangle$ be any normalized shared state on $\mathcal{H}_A \otimes \mathcal{H}_B$. The winning-probability calculation applies to this state as well:

$$\mathbb{P}(\text{win}) = \frac{1}{2} + \frac{1}{8}\langle\psi|M_{AB}|\psi\rangle,$$

where

$$M_{AB} = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1).$$

Each tensor-product summand is Hermitian, so $M_{AB}^\dagger = M_{AB}$. Apply Cauchy–Schwarz to $|\psi\rangle$ and $M_{AB}|\psi\rangle$:

$$\begin{aligned} |\langle\psi|M_{AB}|\psi\rangle|^2 &\le \||\psi\rangle\|^2 \|M_{AB}|\psi\rangle\|^2 \\ &= \langle\psi|M_{AB}^\dagger M_{AB}|\psi\rangle \\ &= \langle\psi|M_{AB}^2|\psi\rangle. \end{aligned}$$

Thus it is enough to prove $\langle\psi|M_{AB}^2|\psi\rangle \le 8$.

Products of Bob's operators

Set $P = B_0 + B_1$ and $Q = B_0 - B_1$. Recall that $(B_0)^2 = (B_1)^2 = I_B$.

Squared terms. Expanding gives

$$P^2 = 2I_B + B_0B_1 + B_1B_0,$$
$$Q^2 = 2I_B - B_0B_1 - B_1B_0,$$

so $P^2 + Q^2 = 4I_B$.

Cross terms. Keep the order of the factors:

$$\begin{aligned} PQ &= (B_0)^2 - B_0B_1 + B_1B_0 - (B_1)^2 \\ &= -B_0B_1 + B_1B_0, \\ QP &= (B_0)^2 + B_0B_1 - B_1B_0 - (B_1)^2 \\ &= B_0B_1 - B_1B_0. \end{aligned}$$

Expanding the square of the CHSH operator

Using $(R \otimes S)(T \otimes U) = RT \otimes SU$, expand $M_{AB} = A_0 \otimes P + A_1 \otimes Q$:

$$\begin{aligned} M_{AB}^2 &= (A_0)^2 \otimes P^2 + (A_1)^2 \otimes Q^2 \\ &\quad + A_0 A_1 \otimes PQ + A_1 A_0 \otimes QP \\ &= I_A \otimes (P^2 + Q^2) \\ &\quad + A_0 A_1 \otimes (-B_0 B_1 + B_1 B_0) \\ &\quad + A_1 A_0 \otimes (B_0 B_1 - B_1 B_0) \\ &= 4 I_A \otimes I_B \\ &\quad - A_0 A_1 \otimes B_0 B_1 + A_0 A_1 \otimes B_1 B_0 \\ &\quad + A_1 A_0 \otimes B_0 B_1 - A_1 A_0 \otimes B_1 B_0. \end{aligned}$$

We will bound the expectation of each of the four remaining products.

Bounding the expectation of a product

For any $x, x', y, y' \in \{0, 1\}$, put

$$U = A_x A_{x'} \otimes B_y B_{y'}.$$

Taking the adjoint reverses the order; the local factors are Hermitian:

$$U^\dagger = A_{x'} A_x \otimes B_{y'} B_y.$$

Since all four local observables square to the identity,

$$\begin{aligned} U^\dagger U &= A_{x'} (A_x)^2 A_{x'} \otimes B_{y'} (B_y)^2 B_{y'} \\ &= (A_{x'})^2 \otimes (B_{y'})^2 = I_A \otimes I_B. \end{aligned}$$

Hence $\|U|\psi\rangle\|^2 = \langle\psi|U^\dagger U|\psi\rangle = 1$. Cauchy–Schwarz gives

$$\boxed{|\langle\psi|U|\psi\rangle| \leq \||\psi\rangle\| \, \|U|\psi\rangle\| = 1.}$$

Bounding the square

Each of the four products in the expansion of M_{AB}^2 has the form U from the previous slide. Also,

$$\langle\psi|M_{AB}^2|\psi\rangle = \|M_{AB}|\psi\rangle\|^2 \geq 0.$$

The triangle inequality therefore gives

$$\begin{aligned}\langle\psi|M_{AB}^2|\psi\rangle &\leq 4 + |\langle\psi|A_0A_1 \otimes B_0B_1|\psi\rangle| \\ &\quad + |\langle\psi|A_0A_1 \otimes B_1B_0|\psi\rangle| \\ &\quad + |\langle\psi|A_1A_0 \otimes B_0B_1|\psi\rangle| \\ &\quad + |\langle\psi|A_1A_0 \otimes B_1B_0|\psi\rangle| \\ &\leq 4 + 1 + 1 + 1 + 1 = 8.\end{aligned}$$

Completion of the Tsirelson bound

Combining the two Cauchy–Schwarz estimates gives

$$\langle\psi|M_{AB}|\psi\rangle \le |\langle\psi|M_{AB}|\psi\rangle| \le \sqrt{\langle\psi|M_{AB}^2|\psi\rangle} \le 2\sqrt{2}.$$

Therefore, for every pure state and projective strategy,

$$\begin{aligned}\mathbb{P}(\text{win}) &= \frac{1}{2} + \frac{1}{8}\langle\psi|M_{AB}|\psi\rangle \\ &\le \frac{1}{2} + \frac{2\sqrt{2}}{8} = \frac{1}{2} + \frac{1}{2\sqrt{2}}.\end{aligned}$$

The reductions extend this to arbitrary states and POVMs. Taking the supremum over all strategies gives the upper bound.
The two-qubit strategy already constructed attains it, so

$$\boxed{\omega_q(G) = \frac{1}{2} + \frac{1}{2\sqrt{2}}.}$$

□

Self-testing a Bell Pair

Self-testing of CHSH

Theorem. For any quantum strategy for the CHSH game that attains the maximum possible success probability of $(1 + 1/\sqrt{2})/2$, there is an orthonormal basis with respect to which Alice and Bob's observables are of the form

$$A_0 = Z \otimes I_m,$$
$$A_1 = X \otimes I_m,$$
$$B_0 = H \otimes I_m,$$
$$B_1 = (ZHZ) \otimes I_m$$

and the entangled state is of the form

$$|\psi\rangle = \left(\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle\right) \otimes |\Phi_{\text{junk}}\rangle$$

for some arbitrary state $|\Phi_{\text{junk}}\rangle \in \mathbb{C}^m \otimes \mathbb{C}^m$.

Setup for the proof

In the main argument, we work with

$$|\psi\rangle = \sum_{j=1}^{d} s_j |j\rangle \otimes |j\rangle, \qquad s_j > 0,$$

and four binary observables satisfying

$$A_x^\dagger = A_x, \quad A_x^2 = I, \qquad B_y^\dagger = B_y, \quad B_y^2 = I.$$

These assumptions are justified in three steps:

1. Full Schmidt rank

Restrict the local spaces to the Schmidt vectors with nonzero coefficients and compress the POVMs accordingly. This preserves all outcome probabilities.

Reducing to an optimal projective strategy

2. Projective measurements

Express each binary POVM as a random mixture of projective measurements on the same space. The shared state remains unchanged and retains full Schmidt rank.

3. Fix the random choices

The original strategy achieves the maximum CHSH score. Since no sampled strategy can exceed that maximum, every sampled strategy with positive probability must also attain it.

We then fix one such choice r and omit the superscripts $A_x^{(r)}, B_y^{(r)}$.

So at this point we are analyzing one optimal projective strategy obtained from the mixture, with the same full-Schmidt-rank state.

Proof: the optimal CHSH value

Let $|\psi\rangle \in \mathbb{C}^d \otimes \mathbb{C}^d$ be a normalized state with full Schmidt rank, and let A_0, A_1, B_0, B_1 be binary observables. Assume the strategy wins CHSH with probability $(1 + 1/\sqrt{2})/2$.

Recall the CHSH operator

$$M_{AB} = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1)$$

and the winning-probability formula

$$\mathbb{P}(\text{win}) = \frac{1}{2} + \frac{1}{8}\langle\psi|M_{AB}|\psi\rangle.$$

Optimality therefore gives

$$\langle\psi|M_{AB}|\psi\rangle = 2\sqrt{2}, \qquad \boxed{\frac{1}{4}\langle\psi|M_{AB}|\psi\rangle = \frac{1}{\sqrt{2}}.}$$

Rewriting optimality with block matrices

Expanding the CHSH operator, the optimality identity is

$$\frac{1}{4}\langle\psi|(A_0 \otimes B_0 + A_0 \otimes B_1 + A_1 \otimes B_0 - A_1 \otimes B_1)|\psi\rangle = \frac{1}{\sqrt{2}}.$$

Equivalently,

$$\langle\psi| \begin{bmatrix} \frac{1}{\sqrt{2}} A_0 \otimes I & \frac{1}{\sqrt{2}} A_1 \otimes I \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} I \otimes \frac{B_0 + B_1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} I \otimes \frac{B_0 - B_1}{\sqrt{2}} \end{bmatrix} |\psi\rangle = 1.$$

The bracketed expressions consist of two square blocks, stacked horizontally or vertically.

Their product is $M_{AB}/(2\sqrt{2})$.

The block matrices are isometries

Put $\mathcal{H} = \mathbb{C}^d \otimes \mathbb{C}^d$. Define $M_A, M_B \colon \mathcal{H} \to \mathcal{H} \oplus \mathcal{H}$ by

$$M_A = \begin{bmatrix} \frac{1}{\sqrt{2}} A_0 \otimes I \\ \frac{1}{\sqrt{2}} A_1 \otimes I \end{bmatrix}, \qquad M_B = \begin{bmatrix} \frac{1}{\sqrt{2}} I \otimes \frac{B_0 + B_1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} I \otimes \frac{B_0 - B_1}{\sqrt{2}} \end{bmatrix}.$$

Since the observables are Hermitian and $A_x^2 = B_y^2 = I$,

$$M_A^\dagger M_A = \frac{1}{2}(A_0^2 + A_1^2) \otimes I = I_{\mathcal{H}},$$

$$M_B^\dagger M_B = \frac{1}{4} I \otimes \left[(B_0 + B_1)^2 + (B_0 - B_1)^2\right] = I_{\mathcal{H}}.$$

Thus both block matrices are isometries.

Optimality gives two identities on the state

Define $|v\rangle = M_A|\psi\rangle$ and $|w\rangle = M_B|\psi\rangle$. Since M_A, M_B are isometries and $|\psi\rangle$ is normalized,

$$\||v\rangle\| = \||w\rangle\| = 1.$$

The block-matrix form of optimality gives

$$\langle v \mid w\rangle = \langle\psi|M_A^\dagger M_B|\psi\rangle = 1.$$

Two unit vectors with inner product 1 are equal, so $|v\rangle = |w\rangle$.

Equating the two blocks and cancelling $1/\sqrt{2}$ yields

$$\boxed{\begin{aligned}(A_0 \otimes I)|\psi\rangle &= \left(I \otimes \frac{B_0 + B_1}{\sqrt{2}}\right)|\psi\rangle, \\ (A_1 \otimes I)|\psi\rangle &= \left(I \otimes \frac{B_0 - B_1}{\sqrt{2}}\right)|\psi\rangle.\end{aligned}}$$

Full Schmidt rank and transposes

Lemma. Let

$$|\psi\rangle = \sum_{j=1}^{d} s_j |j\rangle \otimes |j\rangle, \qquad s_1 \geq \cdots \geq s_d > 0,$$

and let $A, B \in \mathbb{C}^{d\times d}$ be Hermitian. Then

$$(A \otimes I)|\psi\rangle = (I \otimes B)|\psi\rangle \quad \Longrightarrow \quad B = A^{\mathsf{T}},$$

where the transpose is taken in the Schmidt bases.

Proof sketch

Comparing the coefficient of $|i\rangle \otimes |j\rangle$, then interchanging i, j and using Hermiticity, gives

$$s_j A_{ij} = s_i B_{ji}, \qquad s_i A_{ij} = s_j B_{ji}.$$

Multiplying these equations by s_j and s_i, respectively, yields $(s_j^2 - s_i^2)A_{ij} = 0$.

If $s_i \neq s_j$, then $A_{ij} = B_{ji} = 0$. If $s_i = s_j > 0$, the first equation gives $A_{ij} = B_{ji}$. Thus $B = A^{\mathsf{T}}$. □

Alice's observables anticommute

Since $|\psi\rangle$ has full Schmidt rank, the preceding lemma gives

$$\frac{B_0 + B_1}{\sqrt{2}} = A_0^{\mathsf{T}} \qquad \text{and} \qquad \frac{B_0 - B_1}{\sqrt{2}} = A_1^{\mathsf{T}},$$

where the transpose is taken in the Schmidt bases.

Using $B_0^2 = B_1^2 = I$, direct calculation gives

$$\left(\frac{B_0 + B_1}{\sqrt{2}}\right)\left(\frac{B_0 - B_1}{\sqrt{2}}\right) = \frac{B_1 B_0 - B_0 B_1}{2}$$

$$= -\left(\frac{B_0 - B_1}{\sqrt{2}}\right)\left(\frac{B_0 + B_1}{\sqrt{2}}\right).$$

Thus A_0^{T} and A_1^{T} anticommute. Taking transposes shows that A_0 and A_1 anticommute:

$$\boxed{A_0 A_1 = -A_1 A_0.}$$

Anticommuting binary observables

Lemma. If A and B are binary observables on $\mathbb{C}^d$ satisfying $AB = -BA$, then $d = 2m$ is even and there is a unitary U such that

$$U^\dagger A U = Z \otimes I_m, \qquad U^\dagger B U = X \otimes I_m.$$

Proof: block structure

In an eigenbasis of A, write

$$A = \begin{pmatrix} I_{m_1} & 0 \\ 0 & -I_{m_2} \end{pmatrix}, \qquad B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}.$$

The relation $AB = -BA$ implies $B_{11} = B_{22} = 0$. Since B is Hermitian, $B_{21} = B_{12}^\dagger$. Setting $C = B_{12}$ gives

$$B = \begin{pmatrix} 0 & C \\ C^\dagger & 0 \end{pmatrix}, \qquad C \in \mathbb{C}^{m_1 \times m_2}.$$

The canonical change of basis

Since $B^2 = I$, we have $CC^\dagger = I_{m_1}$ and $C^\dagger C = I_{m_2}$. Taking traces gives $m_1 = \mathrm{tr}(CC^\dagger) = \mathrm{tr}(C^\dagger C) = m_2$. Put $m = m_1 = m_2$. Then $d = 2m$ and C is unitary.
Set

$$U_0 = \begin{pmatrix} I_m & 0 \\ 0 & C^\dagger \end{pmatrix}.$$

This is unitary. Using $CC^\dagger = I_m$ gives

$$U_0^\dagger A U_0 = \begin{pmatrix} I_m & 0 \\ 0 & C \end{pmatrix} \begin{pmatrix} I_m & 0 \\ 0 & -I_m \end{pmatrix} \begin{pmatrix} I_m & 0 \\ 0 & C^\dagger \end{pmatrix} = Z \otimes I_m,$$

$$U_0^\dagger B U_0 = \begin{pmatrix} I_m & 0 \\ 0 & C \end{pmatrix} \begin{pmatrix} 0 & C \\ C^\dagger & 0 \end{pmatrix} \begin{pmatrix} I_m & 0 \\ 0 & C^\dagger \end{pmatrix} = X \otimes I_m.$$

Composing U_0 with the initial change to an eigenbasis of A gives the required unitary U. □

Applying the anticommutation lemma

Since A_0 and A_1 anticommute, the lemma gives $d = 2m$ and a unitary U such that

$$U^\dagger A_0 U = Z \otimes I_m, \qquad U^\dagger A_1 U = X \otimes I_m.$$

We also have the transpose identity

$$(U^\dagger A U)^{\mathsf{T}} = U^{\mathsf{T}} A^{\mathsf{T}} (U^{\mathsf{T}})^\dagger.$$

Using $B_0 = (A_0^{\mathsf{T}} + A_1^{\mathsf{T}})/\sqrt{2}$ and $B_1 = (A_0^{\mathsf{T}} - A_1^{\mathsf{T}})/\sqrt{2}$, we obtain

$$U^{\mathsf{T}} B_0 (U^{\mathsf{T}})^\dagger = U^{\mathsf{T}} \frac{A_0^{\mathsf{T}} + A_1^{\mathsf{T}}}{\sqrt{2}} (U^{\mathsf{T}})^\dagger = \frac{Z + X}{\sqrt{2}} \otimes I_m = H \otimes I_m,$$

$$U^{\mathsf{T}} B_1 (U^{\mathsf{T}})^\dagger = U^{\mathsf{T}} \frac{A_0^{\mathsf{T}} - A_1^{\mathsf{T}}}{\sqrt{2}} (U^{\mathsf{T}})^\dagger = \frac{Z - X}{\sqrt{2}} \otimes I_m = (ZHZ) \otimes I_m.$$

Here $Z^{\mathsf{T}} = Z$ and $X^{\mathsf{T}} = X$.

The observables in the new local bases

The local change of coordinates

$$U^\dagger \otimes U^\mathsf{T}$$

puts Alice's and Bob's observables into the form

$$A_0 = Z \otimes I_m,$$

$$A_1 = X \otimes I_m,$$

$$B_0 = H \otimes I_m,$$

$$B_1 = (ZHZ) \otimes I_m.$$

From now on, A_0, A_1, B_0, B_1 denote the observables in these new local bases.

The CHSH operator in the new bases

Write $|\psi\rangle$ for the state in the new local bases. It still satisfies

$$\langle\psi|\frac{A_0\otimes B_0+A_0\otimes B_1+A_1\otimes B_0-A_1\otimes B_1}{4}|\psi\rangle=\frac{1}{\sqrt{2}}.$$

Grouping the two qubit factors together, we have

$$\begin{aligned}\frac{1}{4}M_{AB}&=\frac{Z\otimes Z+X\otimes X}{2\sqrt{2}}\otimes I_{m^2}\\&=\frac{1}{\sqrt{2}}\begin{pmatrix}\frac{1}{2}&0&0&\frac{1}{2}\\0&-\frac{1}{2}&\frac{1}{2}&0\\0&\frac{1}{2}&-\frac{1}{2}&0\\\frac{1}{2}&0&0&\frac{1}{2}\end{pmatrix}\otimes I_{m^2}.\end{aligned}$$

Here $I_{m^2}=I_m\otimes I_m$ acts on the two auxiliary factors $\mathbb{C}^m\otimes\mathbb{C}^m$.

The maximal eigenvalue determines the state

$K = (Z \otimes Z + X \otimes X)/(2\sqrt{2})$ has eigenvalues $1/\sqrt{2}, 0, 0, -1/\sqrt{2}$. Thus $T = M_{AB}/4 = K \otimes I_{m^2}$ has maximum eigenvalue $\lambda_{\max} = 1/\sqrt{2}$.
Expand $|\psi\rangle$ in an orthonormal eigenbasis of T:

$$|\psi\rangle = \sum_r c_r |e_r\rangle, \qquad T|e_r\rangle = \lambda_r |e_r\rangle, \qquad \sum_r |c_r|^2 = 1.$$

$$\begin{aligned}\langle\psi|T|\psi\rangle &= \sum_{r,s} \overline{c_r} c_s \langle e_r|T|e_s\rangle \\ &= \sum_{r,s} \overline{c_r} c_s \lambda_s \underbrace{\langle e_r \mid e_s\rangle}_{\delta_{rs}} = \sum_r \lambda_r |c_r|^2.\end{aligned}$$

Using normalization and optimality,

$$\begin{aligned}0 &= \lambda_{\max} - \langle\psi|T|\psi\rangle = \lambda_{\max} \sum_r |c_r|^2 - \sum_r \lambda_r |c_r|^2 \\ &= \sum_r |c_r|^2 (\lambda_{\max} - \lambda_r).\end{aligned}$$

Every term is nonnegative, so $c_r = 0$ whenever $\lambda_r < \lambda_{\max}$. Therefore $|\psi\rangle$ lies in the maximal eigenspace of T.

The Bell pair and the junk state

The $1/\sqrt{2}$-eigenspace of K is spanned by

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle.$$

Let $\{|f_j\rangle\}_j$ be an orthonormal basis of $\mathbb{C}^m \otimes \mathbb{C}^m$. The maximal eigenspace of $T = K \otimes I_{m^2}$ has basis

$$\left\{|\Phi^+\rangle \otimes |f_j\rangle\right\}_j.$$

Since $|\psi\rangle$ lies in this space,

$$|\psi\rangle = \sum_j d_j |\Phi^+\rangle \otimes |f_j\rangle = |\Phi^+\rangle \otimes \underbrace{\left(\sum_j d_j |f_j\rangle\right)}_{|\Phi_{\text{junk}}\rangle}.$$

The junk state is normalized because $\||\Phi_{\text{junk}}\rangle\|^2 = \sum_j |d_j|^2 = 1$.

This completes the proof of the theorem. □

From the exact theorem to an experiment

Our theorem concerns the true winning probability p. If

$$p = p_* := \frac{1}{2} + \frac{1}{2\sqrt{2}} \approx 0.8536,$$

the shared state contains a Bell pair, up to local isometries and an auxiliary state. We need not know the measurement bases or local dimensions.

In a finite experiment, we observe the fraction of wins

$$\widehat{p} = \frac{\text{number of wins}}{\text{number of rounds}}.$$

This estimates p and has statistical uncertainty. Even a measured fraction very close to p_* cannot establish the exact equality $p = p_*$.

From observed wins to a state guarantee

For independent repetitions of a fixed CHSH strategy, a finite experiment needs both steps, in this order.

1. Statistical analysis: bound the true probability
Use the observed fraction of wins $\widehat{p}$ to obtain a statement such as

$$p \geq p_* - \varepsilon \qquad \text{with 99\% confidence.}$$

2. Robust self-testing: turn this into a state bound
The probability bound guarantees that, after local isometries, the state is within $C\sqrt{\varepsilon}$ of a Bell pair tensor an auxiliary state.

The resulting state guarantee holds with the same 99% confidence.

We state the robust theorem next, without proof.

Robust self-testing of a Bell pair

Suppose a finite-dimensional strategy with pure shared state $|\psi\rangle$ has true CHSH winning probability

$$p \geq p_* - \varepsilon, \qquad \varepsilon \geq 0.$$

Then there exist local isometries V_A, V_B and a normalized auxiliary state $|\xi\rangle$ such that

$$\boxed{\|(V_A \otimes V_B)|\psi\rangle - |\Phi^+\rangle \otimes |\xi\rangle\|_2 \leq C\sqrt{\varepsilon}.}$$

Here $\|\cdot\|_2$ is the Euclidean vector norm, and C is a universal constant, independent of the local dimensions.

A nearly optimal score gives a state close to a Bell pair tensor an auxiliary state. For $\varepsilon = 0$, we recover exact equality.

For mixed states, apply the statement to a purification, with the purifying system left untouched.

The Mermin–Peres Magic Square

A square of signs

Fill the nine cells with signs $s_{ij} \in \{+1, -1\}$. We require every row product to be $+1$ and the column products to be $+1, +1, -1$:

s_{11}	s_{12}	s_{13}	$+1$
s_{21}	s_{22}	s_{23}	$+1$
s_{31}	s_{32}	s_{33}	$+1$
$+1$	$+1$	-1	

Write $c_1 = c_2 = +1$ and $c_3 = -1$ for the column targets.

A solution is a filling satisfying all six product conditions. We turn this puzzle into a game between a referee and two players.

The magic-square game

The referee chooses $x, y \in \{1, 2, 3\}$ uniformly and independently. Alice receives the row index x; Bob receives the column index y. They cannot communicate after receiving their questions.

Alice returns a row $a = (a_1, a_2, a_3)$ and Bob returns a column $b = (b_1, b_2, b_3)$, with entries in $\{+1, -1\}$. They win if

$$\boxed{a_1 a_2 a_3 = +1, \qquad b_1 b_2 b_3 = c_y, \qquad a_y = b_x.}$$

- Alice's row has the required product.
- Bob's column has the required product.
- Their signs agree at the common cell (x, y).

They may agree on a strategy and share resources beforehand.

A filling gives a perfect classical strategy

Suppose $S = (s_{ij})$ solves the square. Alice and Bob agree on this filling before the game.

On input x, Alice returns row x; on input y, Bob returns column y:

$$a_j = s_{xj}, \qquad b_i = s_{iy}.$$

The row and column conditions give

$$\prod_{j=1}^{3} a_j = +1, \qquad \prod_{i=1}^{3} b_i = c_y.$$

At the intersection,

$$a_y = s_{xy} = b_x.$$

Thus every question pair is won, and $\omega_c(G) = 1$.

A perfect classical strategy gives a filling

Suppose a deterministic strategy wins every question pair. Write Alice's answer to x as $a(x)$ and Bob's answer to y as $b(y)$. Define

$$s_{xy} = a_y(x).$$

Alice's winning conditions give the required row products. Agreement at every intersection forces

$$s_{xy} = a_y(x) = b_x(y).$$

Thus column y is precisely Bob's answer to y and has product c_y. So S solves the square.

Randomization cannot improve the optimum, and there are only finitely many deterministic strategies. Thus the classical value is attained by a deterministic strategy, giving

$$\boxed{\omega_c(G) = 1 \quad \Longleftrightarrow \quad \text{the square has a solution.}}$$

The parity contradiction

If a solution existed, multiplying its row products would give

$$\prod_{i,j=1}^{3} s_{ij} = \prod_{i=1}^{3} \left(\prod_{j=1}^{3} s_{ij} \right) = (+1)^3 = +1.$$

Multiplying its column products would give

$$\prod_{i,j=1}^{3} s_{ij} = \prod_{j=1}^{3} \left(\prod_{i=1}^{3} s_{ij} \right) = c_1 c_2 c_3 = -1.$$

Both expressions multiply the same nine numbers: a contradiction. Therefore the square has no solution and $\omega_c(G) < 1$.

In fact, every deterministic strategy loses at least one of the nine equally likely question pairs, so $\omega_c(G) \leq 8/9$. □

An operator solution on two qubits

Replace the signs by observables M_{ij} on $\mathbb{C}^2 \otimes \mathbb{C}^2$. Here $I = I_2$:

$$(M_{ij}) = \begin{pmatrix} Z \otimes I & I \otimes Z & Z \otimes Z \\ I \otimes X & X \otimes I & X \otimes X \\ Z \otimes X & X \otimes Z & Y \otimes Y \end{pmatrix}.$$

Each entry is Hermitian and squares to I_4, since

$$X^2 = Y^2 = Z^2 = I.$$

Its eigenvalues are therefore in $\{+1, -1\}$.

Both players will use this same square. Each entry acts on one player's two qubits.

Commutativity on separate tensor factors

For the first two rows and first two columns, write the entries as A, B, C in their natural order:

	A	B	$C = AB$
Row 1	$Z \otimes I$	$I \otimes Z$	$Z \otimes Z$
Row 2	$I \otimes X$	$X \otimes I$	$X \otimes X$
Column 1	$Z \otimes I$	$I \otimes X$	$Z \otimes X$
Column 2	$I \otimes Z$	$X \otimes I$	$X \otimes Z$

In each case, A and B act on separate tensor factors, so $AB = BA$.

If A and B commute, each also commutes with their product:

$$A(AB) = (AB)A, \qquad B(AB) = (AB)B.$$

Thus all three entries commute. The same argument works if $C = -AB$.

The third row and the third column

Use the Pauli identities $ZX = iY$ and $XZ = -iY$. For the first two entries of the third row,

$$(Z \otimes X)(X \otimes Z) = (iY) \otimes (-iY) = Y \otimes Y,$$
$$(X \otimes Z)(Z \otimes X) = (-iY) \otimes (iY) = Y \otimes Y.$$

They commute, and the third entry is their product.

For the first two entries of the third column,

$$(Z \otimes Z)(X \otimes X) = (iY) \otimes (iY) = -Y \otimes Y,$$
$$(X \otimes X)(Z \otimes Z) = (-iY) \otimes (-iY) = -Y \otimes Y.$$

They commute, and the third entry is minus their product. Hence all three entries commute in these two cases as well.

Commuting observables have a joint eigenbasis

Let A, B, C be the three Hermitian observables in a row or column. Diagonalize A, with eigenspaces E_λ. For $v \in E_\lambda$, commutativity gives

$$A(Bv) = BAv = \lambda Bv.$$

Thus B preserves E_λ. Its restriction is Hermitian: for $u, v \in E_\lambda$,

$$\langle u, Bv \rangle = \langle Bu, v \rangle.$$

Diagonalizing B on each E_λ gives common eigenspaces of A and B.

Since C commutes with both, it preserves each common eigenspace. Diagonalize its Hermitian restriction there.
We obtain an orthonormal basis of simultaneous eigenvectors for A, B, C. □

Products along the rows

In each row, the product of the first two entries is the third. Therefore

$$M_{11}M_{12}M_{13} = (Z \otimes Z)^2 = I_4,$$

$$M_{21}M_{22}M_{23} = (X \otimes X)^2 = I_4,$$

$$M_{31}M_{32}M_{33} = (Y \otimes Y)^2 = I_4.$$

Here, for each Pauli matrix P,

$$(P \otimes P)^2 = P^2 \otimes P^2 = I \otimes I = I_4.$$

Thus every row has the required product $+I_4$.

Products along the columns

In the first two columns, the product of the first two entries is the third. In the last column it is minus the third:

$$M_{11}M_{21}M_{31} = (Z \otimes X)^2 = I_4,$$

$$M_{12}M_{22}M_{32} = (X \otimes Z)^2 = I_4,$$

$$M_{13}M_{23}M_{33} = -(Y \otimes Y)^2 = -I_4.$$

These are precisely the required column products $c_y I_4$.

There is no scalar parity contradiction: entries from different rows and columns need not commute. We cannot freely reorder all nine factors as we did for signs.

The shared state: two Bell pairs

Each player holds two qubits: $\mathcal{H}_A = \mathcal{H}_B = \mathbb{C}^2 \otimes \mathbb{C}^2$. They share the state

$$\begin{aligned}|\psi\rangle &= |\Phi^+\rangle_{A_1B_1} \otimes |\Phi^+\rangle_{A_2B_2} \\ &= \frac{1}{2}(|0\rangle_{A_1} \otimes |0\rangle_{B_1} + |1\rangle_{A_1} \otimes |1\rangle_{B_1}) \\ &\qquad \otimes (|0\rangle_{A_2} \otimes |0\rangle_{B_2} + |1\rangle_{A_2} \otimes |1\rangle_{B_2}).\end{aligned}$$

Alice controls A_1, A_2; Bob controls B_1, B_2.

The tensor factors of each observable $P \otimes Q$ act on a player's first and second qubit, respectively.

One measurement, three signs

Let A, B, C be a row or column of the operator square. Choose a common orthonormal eigenbasis:

$$A|e_k\rangle = \alpha_k|e_k\rangle, \quad B|e_k\rangle = \beta_k|e_k\rangle, \quad C|e_k\rangle = \gamma_k|e_k\rangle.$$

All three eigenvalues are signs in $\{+1, -1\}$.

Perform the projective measurement

$$P_k = |e_k\rangle\langle e_k|, \qquad k = 1, 2, 3, 4, \qquad \sum_{k=1}^{4} P_k = I_4.$$

On outcome k, return $(\alpha_k, \beta_k, \gamma_k)$.

This is how we measure the three commuting observables together: the three signs label one measurement outcome.

Alice's and Bob's measurements

- On question x, Alice measures a common eigenbasis of M_{x1}, M_{x2}, M_{x3} and returns their eigenvalues a_1, a_2, a_3.
- On question y, Bob measures a common eigenbasis of M_{1y}, M_{2y}, M_{3y} and returns their eigenvalues b_1, b_2, b_3.

Example: Alice receives $x = 3$

She jointly measures

$$Z \otimes X, \qquad X \otimes Z, \qquad Y \otimes Y.$$

If the basis outcome is $|u\rangle$, then

$$(Z \otimes X)|u\rangle = a_1|u\rangle, \quad (X \otimes Z)|u\rangle = a_2|u\rangle, \quad (Y \otimes Y)|u\rangle = a_3|u\rangle.$$

Each player's choice depends only on their own question.

The products of the answers are correct

Suppose A, B, C is a measured row or column with

$$ABC = \varepsilon I_4, \qquad \varepsilon \in \{+1, -1\}.$$

For any common basis outcome $|e\rangle$ with eigenvalues α, β, γ,

$$\varepsilon|e\rangle = ABC|e\rangle = \alpha\beta\gamma|e\rangle.$$

Since $|e\rangle \neq 0$, the reported signs satisfy $\alpha\beta\gamma = \varepsilon$.

Applying the row and column products proved above, we get with certainty

$$\boxed{a_1 a_2 a_3 = +1, \qquad b_1 b_2 b_3 = c_y.}$$

This holds for every state being measured. It remains to prove agreement at the intersection: $a_y = b_x$.

Three elementary Bell-pair identities

For $|\Phi^+\rangle = (|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)/\sqrt{2}$, applying the Pauli matrices to the two qubits gives

$$(X \otimes X)|\Phi^+\rangle = \frac{|1\rangle \otimes |1\rangle + |0\rangle \otimes |0\rangle}{\sqrt{2}} = |\Phi^+\rangle,$$

$$(Z \otimes Z)|\Phi^+\rangle = \frac{|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle}{\sqrt{2}} = |\Phi^+\rangle,$$

$$(Y \otimes Y)|\Phi^+\rangle = \frac{-|1\rangle \otimes |1\rangle - |0\rangle \otimes |0\rangle}{\sqrt{2}} = -|\Phi^+\rangle.$$

For the last line, recall $Y|0\rangle = i|1\rangle$ and $Y|1\rangle = -i|0\rangle$.

Also $(I \otimes I)|\Phi^+\rangle = |\Phi^+\rangle$. Applying the same Pauli matrix to both halves preserves the Bell pair, with a minus sign only for Y.

The shared state and a common cell

Let $M = P \otimes Q$ be any cell of the square. To compute $(M \otimes M)|\psi\rangle$, group the four qubits by their two Bell pairs:

$$(M \otimes M)|\psi\rangle = \left((P \otimes P)|\Phi^+\rangle\right)_{A_1B_1} \otimes \left((Q \otimes Q)|\Phi^+\rangle\right)_{A_2B_2}.$$

The first pair gets $P \otimes P$; the second gets $Q \otimes Q$.

- Every cell except $Y \otimes Y$ uses only I, X, Z. Both Bell pairs are therefore unchanged.
- For $M = Y \otimes Y$, both Bell pairs change sign, and the two signs cancel:

$$\left(-|\Phi^+\rangle\right)_{A_1B_1} \otimes \left(-|\Phi^+\rangle\right)_{A_2B_2} = |\psi\rangle.$$

Hence, for every cell M,

$$\boxed{(M \otimes M)|\psi\rangle = |\psi\rangle.}$$

Why the intersection answers agree

Fix x, y and $M = M_{xy}$. Alice's and Bob's basis outcomes $|u\rangle, |v\rangle$ satisfy

$$M|u\rangle = a_y|u\rangle, \qquad M|v\rangle = b_x|v\rangle.$$

Their product basis vector $|w\rangle = |u\rangle \otimes |v\rangle$ satisfies

$$(M \otimes M)|w\rangle = a_y b_x |w\rangle.$$

If the signs disagree, this eigenvalue is -1, whereas $|\psi\rangle$ has eigenvalue $+1$ for the same Hermitian operator. These eigenvectors are orthogonal, so the Born rule gives

$$\mathbb{P}(\text{outcome } u, v \mid x, y) = |\langle w \mid \psi\rangle|^2 = 0.$$

Thus the intersection signs always agree. The row and column products are also correct: the strategy wins every question pair. Therefore

$$\boxed{\omega_c(G) < 1 = \omega_q(G).}$$

Quantum Contextuality

Classical statements are sets of outcomes

Consider two labelled coins, with heads H and tails T:

$$\Omega = \{\mathrm{HH}, \mathrm{HT}, \mathrm{TH}, \mathrm{TT}\}.$$

A statement corresponds to the outcomes where it is true:

$$A = \{\mathrm{HH}, \mathrm{HT}\} \qquad \text{“the first coin is heads”},$$
$$B = \{\mathrm{HH}, \mathrm{TH}\} \qquad \text{“the second coin is heads”}.$$

Combining statements gives

$A \wedge B$	both coins are heads	$\{\mathrm{HH}\}$
$A \vee B$	at least one is heads	$\{\mathrm{HH}, \mathrm{HT}, \mathrm{TH}\}$
$\neg A$	the first coin is tails	$\{\mathrm{TH}, \mathrm{TT}\}$

All subsets of Ω form the Boolean algebra $\mathcal{P}(\Omega)$, with intersection, union and complement as “and”, “or” and “not”.

A classical outcome gives all truth values

Each possible outcome $\omega \in \Omega$ determines a truth assignment to every event $E \subseteq \Omega$:

$$v_\omega(E) = \begin{cases} 1, & \omega \in E, \\ 0, & \omega \notin E. \end{cases}$$

For example, if $\omega = \mathrm{HT}$, the statement "the first coin is heads" is true and "the second coin is heads" is false:

$$v_{\mathrm{HT}}(A) = 1, \qquad v_{\mathrm{HT}}(B) = 0.$$

The compound statements have the expected truth values:

$$v_{\mathrm{HT}}(A \wedge B) = 0, \qquad v_{\mathrm{HT}}(A \vee B) = 1.$$

These values are consistent for all statements at once.

A classical probability model assigns probabilities to these possible outcomes, hence to these truth assignments.

Qubits and measurement statements

A qubit has state space $\mathbb{C}^2$. Choosing an observable determines which basis we measure in and how we label its outcomes:

Observable	$+1$ eigenstate	-1 eigenstate
Z	$\lvert 0\rangle$	$\lvert 1\rangle$
X	$\lvert +\rangle$	$\lvert -\rangle$

Here

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

The statement "a Z measurement gives $+1$" corresponds to the projector

$$P_{Z,+} = |0\rangle\langle 0| = \frac{I + Z}{2}.$$

For a state ρ, its probability is $\mathrm{tr}(\rho P_{Z,+})$.

What does $X \otimes Z$ measure?

Take two qubits. Measure X on the first and Z on the second, obtaining outcomes $a, b \in \{+1, -1\}$. These are the outcomes of $X \otimes I$ and $I \otimes Z$.

Reporting their product ab gives the outcome statistics of $X \otimes Z$:

Qubit 1: X	Qubit 2: Z	$X \otimes Z$
$+1$	$+1$	$+1$
-1	-1	$+1$
$+1$	-1	-1
-1	$+1$	-1

The statement "$X \otimes Z$ has outcome $+1$" means

the two measurement outcomes agree: $a = b$.

Both the $++$ and the $--$ outcomes make this statement true.

The statement "the outcomes agree"

Let P mean "X on qubit 1 gives $+1$", and Q mean "Z on qubit 2 gives $+1$":

$$P = \frac{I + X}{2} \otimes I = |+\rangle\langle+| \otimes I, \qquad Q = I \otimes \frac{I + Z}{2} = I \otimes |0\rangle\langle 0|.$$

The projectors commute: $PQ = QP$.

"Both outcomes are $+1$"

$$P \wedge Q = PQ = |+\rangle\langle+| \otimes |0\rangle\langle 0|.$$

"The outcomes agree"

This means both are $+1$ *or* both are -1:

$$\begin{aligned} R &= (P \wedge Q) \vee (\neg P \wedge \neg Q) \\ &= PQ + (I_4 - P)(I_4 - Q) \\ &= \frac{I_4 + X \otimes Z}{2}. \end{aligned}$$

Thus R is precisely the projector for outcome $+1$ of $X \otimes Z$.

The same square, six measurement contexts

On two qubits, use these Pauli observables and their products:

$Z \otimes I$	$I \otimes Z$	$Z \otimes Z$	$+I_4$
$I \otimes X$	$X \otimes I$	$X \otimes X$	$+I_4$
$Z \otimes X$	$X \otimes Z$	$Y \otimes Y$	$+I_4$
$+I_4$	$+I_4$	$-I_4$	

A measurement context is a family of compatible observables. Each row and column is a context, since its observables commute and can be measured together.

Each observable occurs in two contexts. For example, $Z \otimes I$ occurs in the first row and the first column.

A noncontextual value assignment assigns each observable a fixed value ± 1, independent of the row or column in which it is measured. These values must satisfy every row and column product constraint.

The same qubit statement in two contexts

The proposition "the two outcomes agree" is

$$R = \frac{I_4 + X \otimes Z}{2}.$$

In the second column

Jointly measure

$$I \otimes Z, \qquad X \otimes I, \qquad X \otimes Z.$$

Read the individual outcomes and multiply them.

In the third row

Jointly measure the three commuting products

$$Z \otimes X, \qquad X \otimes Z, \qquad Y \otimes Y$$

in a common eigenbasis. The single-qubit observables X, Y, Z remain incompatible.

The same proposition R occurs in both contexts. A noncontextual assignment must give it the same truth value.

Yes/no questions and Boolean contexts

A quantum yes/no proposition is a projector $P = P^\dagger = P^2$. Its complement, $I - P$, represents the answer "no".

For commuting projectors P, Q, define

$$\neg P = I - P, \qquad P \wedge Q = PQ, \qquad P \vee Q = P + Q - PQ.$$

These are the Boolean operations "not", "and", and "or".

For X on the first qubit and Z on the second, a common basis is $|+\rangle \otimes |0\rangle, |+\rangle \otimes |1\rangle, |-\rangle \otimes |0\rangle, |-\rangle \otimes |1\rangle$. Label these $|e_1\rangle, \ldots, |e_4\rangle$. For $S \subseteq \{1, 2, 3, 4\}$, put

$$E_S = \sum_{k \in S} |e_k\rangle\langle e_k|.$$

Then

$$E_S \wedge E_T = E_{S \cap T}, \qquad E_S \vee E_T = E_{S \cup T}, \qquad \neg E_S = E_{S^c}.$$

Thus propositions within a context obey ordinary Boolean logic.

Partial Boolean algebras

A partial Boolean algebra consists of a set $\mathcal{A}$ with

- a reflexive, symmetric compatibility relation $\sim$;
- constants $0, 1$ and a complement $\neg a$ for every a;
- operations $a \wedge b$ and $a \vee b$, defined when $a \sim b$.

The defining requirement is:

Every pairwise compatible family lies in a pairwise compatible subset that forms a Boolean algebra under these operations.

The quantum example

Take all projectors on $\mathcal{H}$, with

$$P \sim Q \quad \Longleftrightarrow \quad PQ = QP, \qquad 0 = 0_{\mathcal{H}}, \quad 1 = I_{\mathcal{H}}.$$

For example, "X on qubit 1 gives $+1$" and "Z on qubit 1 gives $+1$" have noncommuting projectors. Their Boolean conjunction is undefined in this partial algebra.

Noncontextual truth assignments

A truth assignment is a map $v\colon \mathcal{A} \to \{0,1\}$ preserving the Boolean operations:

$$v(0) = 0, \qquad v(1) = 1, \qquad v(\neg P) = 1 - v(P),$$

and, whenever $P \sim Q$,

$$v(P \wedge Q) = v(P)v(Q), \qquad v(P \vee Q) = \max\{v(P), v(Q)\}.$$

In a joint eigenbasis

The projectors $E_k = |e_k\rangle\langle e_k|$ satisfy $E_k \wedge E_\ell = 0$ for $k \neq \ell$ and $E_1 \vee \cdots \vee E_d = I$. Therefore exactly one E_k has value 1. Thus v selects one joint measurement outcome.

Across contexts

A projector shared by two contexts must have the same truth value in both. The assignments on the individual Boolean algebras must agree wherever they overlap.

Classical and quantum truth assignments

Classical logic

Statements belong to one Boolean algebra. Values assigned to its generating statements, respecting their relations, determine the truth values of all compound statements.

Quantum logic

Each compatible context gives a Boolean algebra. These algebras overlap to form a partial Boolean algebra. A truth assignment is possible within each context. A global assignment must agree on every shared proposition.

The nine statements in the square

For each observable M_{ij}, the projector

$$P_{ij} = \frac{I_4 + M_{ij}}{2}$$

represents "M_{ij} gives $+1$". The square asks whether nine truth values can satisfy the Boolean relations in all six contexts at once.

The square forbids a global truth assignment

Suppose such a v exists on the projectors of $\mathbb{C}^4$. For each observable M_{ij} in the square, set

$$P_{ij} = \frac{I_4 + M_{ij}}{2}, \qquad s_{ij} = 2v(P_{ij}) - 1 \in \{+1, -1\}.$$

Here P_{ij} means "the outcome of M_{ij} is $+1$". The sign s_{ij} is independent of the chosen row or column.

In each context, v selects a joint outcome, so its signs satisfy the operator-product constraint. With $(c_1, c_2, c_3) = (+1, +1, -1)$,

$$\prod_{j=1}^{3} s_{ij} = +1, \qquad \prod_{i=1}^{3} s_{ij} = c_j.$$

Multiplying all six equations counts every sign twice:

$$1 = \prod_{i,j} s_{ij}^2 = (+1)^3(+1)(+1)(-1) = -1.$$

No global truth assignment exists. □

What the Mermin–Peres square demonstrates

There exist partial Boolean algebras with no global truth assignment.

A concrete quantum example

The Mermin–Peres square realizes this obstruction using nine observables on two qubits. The propositions are their associated projectors and Boolean combinations within compatible contexts.

Within each context

Each of the six contexts gives a Boolean algebra admitting ordinary truth assignments.

Across contexts

These assignments cannot be chosen to agree on every shared projector. There is no global map $v : \mathcal{A} \to \{0, 1\}$ preserving the Boolean operations in every context.

Conclusions

CHSH gives a quantum advantage

$$\omega_c(G_{\mathrm{CHSH}}) = \frac{3}{4} < \omega_q(G_{\mathrm{CHSH}}) = \frac{1}{2} + \frac{1}{2\sqrt{2}}.$$

An optimal CHSH score self-tests a Bell pair

Local isometries identify a Bell pair tensor an auxiliary state.

The magic-square game has a perfect quantum strategy

$$\omega_c(G_{\mathrm{MS}}) \leq \frac{8}{9} < \omega_q(G_{\mathrm{MS}}) = 1.$$

The square also proves contextuality

No consistent truth assignment exists across its six contexts, for any quantum state.

Conclusions

Possibility	**Ruled out?**	**Reason**
Noncontextual predetermined values for all nine observables	**Yes**	Cannot satisfy all six quantum product constraints.
Contextual deterministic models	No	Values may depend on the measurement context.
Bell-local models of the perfect game correlations	**Yes**	With uniform inputs independent of hidden variables, success is at most 8/9.
All hidden-variable theories	No	Contextual, nonlocal theories remain possible.
Some definite properties before measurement	No	Some properties can have definite values without a global assignment.
The no-signalling principle	No	The quantum correlations do not enable faster-than-light communication.
An objective physical reality	No	The result does not imply that reality depends on observers.